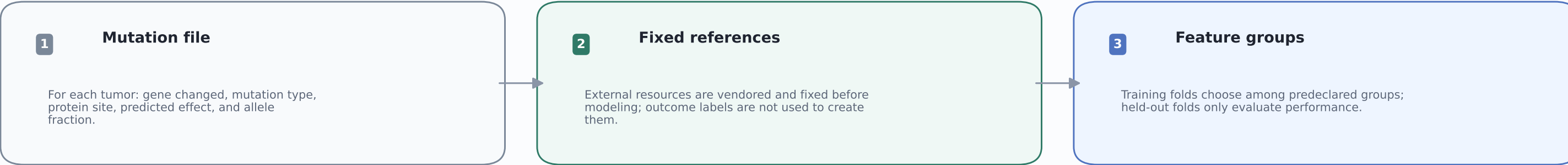


Supplementary Figure S4A. Bio MAF v4 Feature Groups, Without the Jargon

Bio MAF v4 converts an annotated tumor mutation file into fixed, predeclared feature groups. Table S5 gives worked examples, formulas, and definitions.

Strict Bio MAF core: 943 features. Optional controls add 5, for 948 possible Bio MAF features.



Feature group	Count derivation	Example feature	Worked example: toy MAF -> feature value
Mutation-effect and allele-fraction basics 121 features	100 = 50 fixed VEP labels x 2 values; +5 allele-fraction summaries; +12 OncoKB role summaries; +4 non-OncoKB cancer-gene summaries.	<code>`vep_consequence_count__missense_variant`</code>	Toy tumor: 3 missense mutations among 7 annotated mutations. Count = $\log_1p(3) = 1.386$. Paired fraction = $3 / 7 = 0.429$.
External evidence-confidence controls 16 features	16 = 4 groups x 4 measures. Groups are genes named by 1, 2, 3, or 4+ external cancer references.	<code>`driver_evidence_high_or_moderate_impact_log_count__source_count_4`</code>	Toy tumor: one TP53 and one BRCA1 high/moderate-impact mutation. Both genes are in the 4-source group. Value = $\log_1p(2) = 1.099$.
Fixed cancer-gene identity 802 features	802 = 401 externally fixed cancer genes x 2 log-counts: protein-affecting + cancer-role-fit.	<code>`driver_gene_functional_event_log_count__TP53`</code> <code>`driver_gene_role_matched_event_log_count__TP53`</code>	Toy tumor: one TP53 frameshift and one TP53 synonymous mutation. Frameshift counts; synonymous does not. Each TP53 example value = $\log_1p(1) = 0.693$.
Known recurrent cancer sites 4 features	4 hotspot measures.	<code>`cancer_hotspot_residue_log_count`</code> <code>`cancer_hotspot_max_vaf`</code>	Toy tumor: two recurrent hotspot-residue hits with allele fractions 0.21 and 0.47. Count = $\log_1p(2) = 1.099$; max allele fraction = 0.47.
Optional mutation-count controls 5 features	5 mutation-load and event-mix controls.	<code>`log10_id_burden`</code> <code>`id_fraction`</code>	Toy tumor: 80 single-base, 2 double-base, and 18 insertion/deletion events. Burden = $\log_{10}(18 + 1) = 1.279$; fraction = $18 / 100 = 0.180$.

Jargon decoder					
VEP label	Allele fraction	Log-count	Protein-affecting	Cancer-role-fit	Fixed before training
Standard Ensembl effect term assigned to a mutation.	Share of tumor DNA reads carrying the mutation.	$\log_1p(\text{number of matching mutations})$, not an average.	VEP rules say the mutation likely changes/disrupts protein.	Mutation type fits the gene's known cancer role.	Created before model training; no endpoint labels used.